Experiment 16: Adaptive Detection Threshold Analysis

Threshold Sensitivity for Multipath Detection in a 5.9 GHz OFDM-Based V2X-ISAC Channel

V2X–ISAC Research

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Abstract

This experiment investigates the influence of detection threshold selection on OFDM-based multipath delay estimation under different signal-to-noise ratio (SNR) conditions.

A multipath channel generated using NVIDIA Sionna RT at a carrier frequency of 5.9 GHz is used as the simulation reference. Three fixed detection thresholds, -30 dB, -25 dB and -20 dB, are evaluated across six SNR conditions ranging from -5 dB to 20 dB.

For each SNR and threshold combination, 100 Monte Carlo trials are conducted. Performance is evaluated using detection probability, precision, recall, F1-score, false peaks per trial, delay mean absolute error (MAE), and delay root mean square error (RMSE).

The results demonstrate a clear trade-off between weak multipath component detection and false detection suppression. At 20 dB SNR, the -30 dB threshold achieves 100% detection probability and an F1-score of 0.996, whereas the -20 dB threshold achieves 100% precision but detects only 60% of the reference peak opportunities.

The findings indicate that a single fixed threshold does not provide optimal detection performance across all SNR conditions. The results motivate the development of an SNR-dependent adaptive detection threshold.

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1 Introduction

Experiment 15 established the influence of AWGN on multipath detection and delay estimation.

The results demonstrated that detection performance depends on the received SNR. However, the previous experiment used a fixed detection threshold of -25 dB.

A fixed threshold can create two problems:

1. A low threshold can detect weak multipath components but may also identify noise-generated local maxima as valid propagation components.
2. A high threshold can suppress false detections but may reject weaker propagation components.

Experiment 16 investigates this threshold sensitivity by evaluating three different detection thresholds under identical SNR conditions.

The main objectives are:

1. Evaluate the influence of detection threshold selection on multipath detection.
2. Quantify the trade-off between detection probability and false detections.
3. Investigate the recovery of weaker multipath components.
4. Compare delay estimation accuracy under different threshold conditions.
5. Identify a suitable threshold-selection strategy for subsequent adaptive detection.

2 Simulation Configuration

The experiment uses the same Sionna RT-derived multipath channel and OFDM configuration established in the previous experiments.

Table 1: Experiment 16 simulation parameters

Parameter	**Value**
Ray-tracing environment	Simple street canyon
Backend	LLVM
Carrier frequency	5.9 GHz
Sampling frequency	100 MHz
FFT size	1024
Modulation	QPSK
Cyclic prefix	128 samples
Subcarrier spacing	97.65625 kHz
Useful symbol duration	10.24 μs
Valid propagation ray entries	12
Discrete channel length	60 samples
Noise model	Complex AWGN
SNR conditions	-5, 0, 5, 10, 15, 20 dB
Detection thresholds	-30, -25, -20 dB
Monte Carlo trials	100 per condition
Peak matching tolerance	10 ns

A total of:

$$N_{\text{conditions}} = 6 \times 3 = 18 \tag{1}$$

SNR-threshold combinations are evaluated.
The total number of Monte Carlo trials is:

$$N_{\text{total}} = 18 \times 100 = 1800 \tag{2}$$

The same noisy signal realization is evaluated using all three thresholds within each SNR trial. This provides a paired comparison of threshold performance.

3 Reference Multipath Components

Five prominent reference peaks are retained from Experiment 15.
The reference positions are not recalculated when the detection threshold changes.

Table 2: Noise-free reference multipath peaks

Peak	**Reference delay (ns)**	**Magnitude (dB)**
1	131.150812	-4.271
2	149.600922	0.000
3	176.494451	-16.015
4	230.029754	-24.807
5	281.620815	-22.940

The five peaks are used as fixed reference delay positions for evaluating the detection algorithm.

They represent prominent peaks of the sampled noise-free delay profile and should not be interpreted as five independently resolved physical rays.

4 Mathematical Methodology

4.1 AWGN Model

The received signal is represented as:

$$y[n] = h[n] * x[n] + w[n] \tag{3}$$

where $x[n]$ is the transmitted OFDM signal, $h[n]$ is the discrete channel response, and $w[n]$ is complex additive white Gaussian noise.

The noise variance is calculated using:

$$\sigma_w^2 = \frac{P_y}{10^{\mathrm{SNR_{dB}}/10}} \tag{4}$$

where P_y is the average received signal power.

The measured received signal power is:

$$P_y = 1.258177 \times 10^{-11} \tag{5}$$

For example, at 10 dB SNR:

$$\sigma_w^2 = \frac{1.258177 \times 10^{-11}}{10^{10/10}} \tag{6}$$

$$= 1.258177 \times 10^{-12} \tag{7}$$

The complex noise samples are generated as:

$$w[n] = \sqrt{\frac{\sigma_w^2}{2}}\,(w_I[n] + j w_Q[n]) \tag{8}$$

where $w_I[n]$ and $w_Q[n]$ are independent standard Gaussian random variables.

4.2 Delay-Domain Processing

The estimated channel frequency response is:

$$\widehat{H}[k] = \frac{Y[k]}{X[k]} \tag{9}$$

The corresponding delay-domain response is obtained using the inverse discrete Fourier transform:

$$\widehat{h}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \widehat{H}[k] e^{j2\pi kn/N} \tag{10}$$

The magnitude profile is normalized relative to the maximum magnitude of the noise-free reference:

$$A_{\text{norm}}[n] = \frac{|\widehat{h}[n]|}{\max_n |h_{\text{ref}}[n]|} \tag{11}$$

The corresponding magnitude in decibels is:

$$A_{\text{dB}}[n] = 20 \log_{10} \left(\max(A_{\text{norm}}[n], 10^{-12}) \right) \tag{12}$$

This common normalization ensures that the three thresholds are evaluated against the same reference amplitude scale.

4.3 Threshold-Based Peak Detection

A local maximum is accepted when it satisfies:

$$A_{\text{dB}}[n] > A_{\text{dB}}[n-1] \tag{13}$$

$$A_{\text{dB}}[n] \geq A_{\text{dB}}[n+1] \tag{14}$$

and:

$$A_{\text{dB}}[n] \geq T \tag{15}$$

where T is the selected detection threshold.
The evaluated thresholds are:

$$T \in \{-30, -25, -20\} \text{ dB} \tag{16}$$

A lower threshold accepts weaker local maxima, whereas a higher threshold is more selective.

4.4 Fractional-Delay Estimation

The sampling interval is:

$$T_s = \frac{1}{f_s} = \frac{1}{100 \times 10^6} = 10 \text{ ns} \tag{17}$$

Parabolic interpolation is applied to the three samples surrounding a detected peak. Let y_{-1}, y_0, and y_{+1} denote the log-magnitude values surrounding the local maximum. The fractional offset is:

$$\delta = \frac{1}{2} \frac{y_{-1} - y_{+1}}{y_{-1} - 2y_0 + y_{+1}} \tag{18}$$

The estimated delay is:

$$\boxed{\widehat{\tau} = (n_{\text{peak}} + \delta) T_s} \tag{19}$$

The interpolation offset is constrained to:

$$-0.5 \leq \delta \leq 0.5 \tag{20}$$

This provides a sub-sample estimate of the detected peak position.

4.5 One-to-One Peak Matching

A detected peak is considered a match when its delay is within 10 ns of a reference peak:

$$|\widehat{\tau}_j - \tau_i| \leq 10 \text{ ns} \tag{21}$$

Each reference peak and detected peak can be matched only once.
The matching procedure produces:

- True Positives (TP): correctly matched peaks.
- False Positives (FP): unmatched detected peaks.
- False Negatives (FN): undetected reference peaks.

The number of reference peak opportunities per threshold and SNR condition is:

$$N_{\text{reference}} = 100 \times 5 = 500 \tag{22}$$

5 Performance Metrics

5.1 Detection Probability

$$P_D = \frac{TP}{TP + FN} \times 100 \tag{23}$$

5.2 Precision

$$\text{Precision} = \frac{TP}{TP + FP} \times 100 \tag{24}$$

5.3 Recall

$$\text{Recall} = \frac{TP}{TP + FN} \times 100 \tag{25}$$

5.4 F1-score

$$F_1 = 2\frac{P \times R}{P + R} \tag{26}$$

where P and R are precision and recall expressed as fractions.

5.5 False Peaks per Trial

$$N_{\text{FP,avg}} = \frac{FP}{N_{\text{trials}}} \tag{27}$$

5.6 Delay MAE

$$\text{MAE} = \frac{1}{M} \sum_{i=1}^{M} |\widehat{\tau}_i - \tau_i| \tag{28}$$

5.7 Delay RMSE

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(\widehat{\tau}_i - \tau_i)^2} \tag{29}$$

Here, M is the number of matched peak estimates. Consequently, the reported delay errors are conditional on successful detection and matching.

6 Experimental Results

6.1 Detection Performance

Table 3 presents the measured performance for all 18 threshold and SNR combinations.

Table 3: Detection performance under different thresholds

SNR (dB)	Threshold (dB)	Detection (%)	Precision (%)	Recall (%)	F1	FP/trial
-5	-30	85.80	33.81	85.80	0.485	8.40
-5	-25	84.40	35.08	84.40	0.496	7.81
-5	-20	74.60	48.89	74.60	0.591	3.90
0	-30	87.00	37.31	87.00	0.522	7.31
0	-25	84.20	47.73	84.20	0.609	4.61
0	-20	61.00	83.79	61.00	0.706	0.59
5	-30	92.20	51.11	92.20	0.658	4.41
5	-25	84.40	78.44	84.40	0.813	1.16
5	-20	55.60	97.89	55.60	0.709	0.06
10	-30	99.00	75.80	99.00	0.859	1.58
10	-25	91.00	92.48	91.00	0.917	0.37
10	-20	59.00	99.66	59.00	0.741	0.01
15	-30	99.60	91.88	99.60	0.956	0.44
15	-25	91.60	99.35	91.60	0.953	0.03
15	-20	59.60	100.00	59.60	0.747	0.00
20	-30	100.00	99.21	100.00	0.996	0.04
20	-25	92.80	100.00	92.80	0.963	0.00
20	-20	60.00	100.00	60.00	0.750	0.00

6.2 Mathematical Verification at 20 dB

For the -30 dB threshold, the aggregate counts are:

$$TP = 500, \quad FP = 4, \quad FN = 0 \tag{30}$$

Detection probability:

$$P_D = \frac{500}{500+0} \times 100 \tag{31}$$
$$= 100\% \tag{32}$$

Precision:

$$P = \frac{500}{500+4} \times 100 \tag{33}$$
$$= 99.2063\% \tag{34}$$

F1-score:

$$F_1 = 2\frac{1.0000 \times 0.992063}{1.0000 + 0.992063} \tag{35}$$
$$= 0.9960 \tag{36}$$

False peaks per trial:

$$N_{\text{FP,avg}} = \frac{4}{100} = 0.04 \tag{37}$$

For the -20 dB threshold:

$$TP = 300, \quad FP = 0, \quad FN = 200 \tag{38}$$

Therefore:

$$P_D = \frac{300}{500} \times 100 = 60\% \tag{39}$$
$$P = \frac{300}{300} \times 100 = 100\% \tag{40}$$
$$R = \frac{300}{500} \times 100 = 60\% \tag{41}$$

The F1-score becomes:

$$F_1 = 2\frac{1.0 \times 0.60}{1.0 + 0.60} = 0.75 \tag{42}$$

This comparison demonstrates that 100% precision does not necessarily imply complete multipath detection.

6.3 Delay Estimation Accuracy

Table 4: Delay RMSE under different threshold conditions

SNR (dB)	-30 dB (ns)	-25 dB (ns)	-20 dB (ns)
-5	3.3992	3.3343	3.1624
0	3.0233	2.9914	1.7348
5	2.2336	2.0300	0.6514
10	1.1835	1.0246	0.4933
15	0.6056	0.5584	0.2787
20	0.3674	0.3538	0.1591

At 20 dB, the -20 dB threshold produces the smallest RMSE:

$$\mathrm{RMSE}_{-20} = 0.1591 \text{ ns} \tag{43}$$

However, this result is based on only 300 matched peak estimates, compared with 500 matched estimates for the -30 dB threshold.

The higher threshold rejects weaker components. Therefore, the smaller RMSE must not be interpreted as superior overall detection performance.

It is a conditional accuracy result for the components that were successfully detected.

7 Individual Multipath Detection Analysis

The detection probability of each reference peak was calculated independently.

Table 5 presents the results at 20 dB SNR.

Table 5: Individual peak detection probability at 20 dB

Peak	Delay (ns)	-30 dB	-25 dB	-20 dB
1	131.15	100%	100%	100%
2	149.60	100%	100%	100%
3	176.49	100%	100%	100%
4	230.03	100%	64%	0%
5	281.62	100%	100%	0%

The first three reference peaks are detected in all trials at 20 dB for all three thresholds.

Peak 4 and Peak 5 demonstrate the effect of threshold selection.

The -20 dB threshold completely rejects both components at 20 dB, while the -30 dB threshold detects all five reference peaks.

This indicates that increasing the detection threshold can eliminate weak components even when the SNR is high.

7.1 Threshold Margin of the Weakest Peak

Peak 4 has a reference magnitude of approximately -24.807 dB.

For the -25 dB threshold, its threshold margin is:

$$\Delta T = -24.807 - (-25) \tag{44}$$
$$= 0.193 \text{ dB} \tag{45}$$

For the -20 dB threshold:

$$\Delta T = -24.807 - (-20) \tag{46}$$
$$= -4.807 \text{ dB} \tag{47}$$

Therefore, Peak 4 lies below the -20 dB threshold and is rejected by the detector.

Peak 5 has a reference magnitude of -22.940 dB, which also lies below the -20 dB threshold.

These amplitude relationships explain the observed rejection of weaker components.

8 Graphical Results

8.1 Threshold Sensitivity Analysis

Figure 1 presents the four principal performance metrics across the tested SNR and threshold conditions.

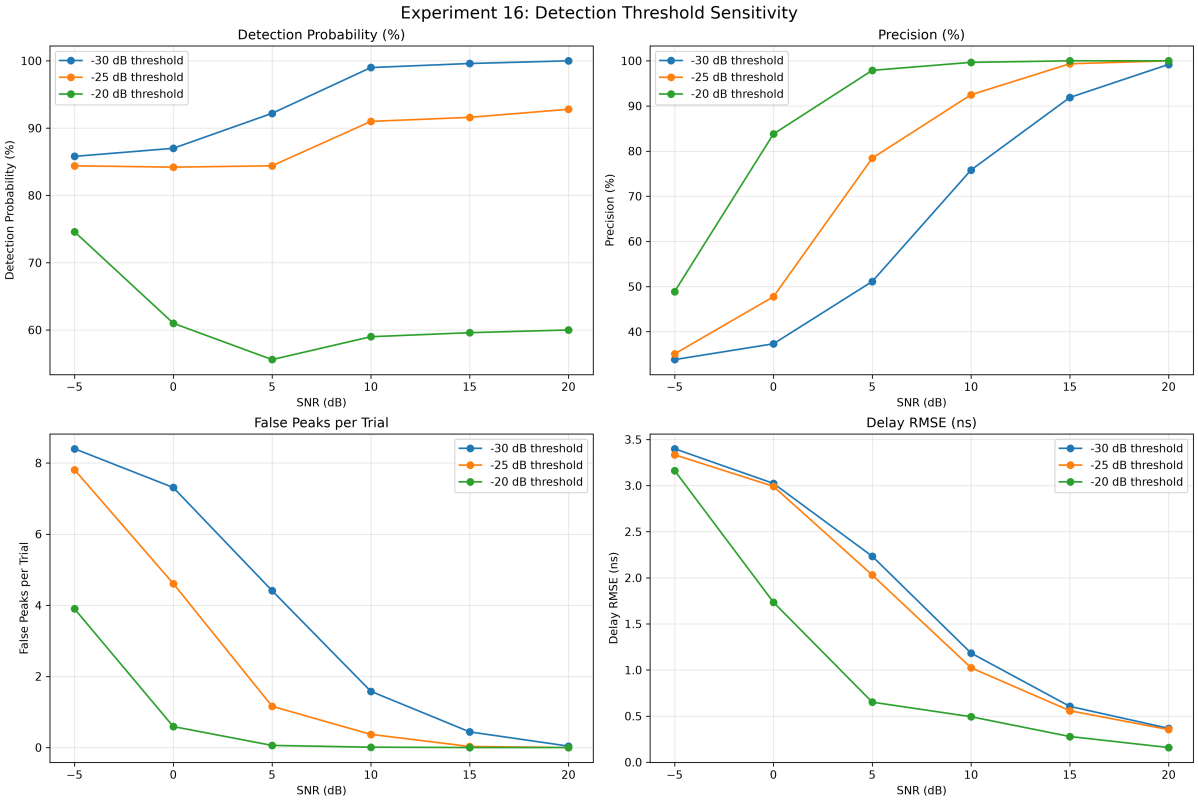


Figure 1: Experiment 16 threshold sensitivity analysis: detection probability, precision, false peaks per trial and delay RMSE.

The detection probability graph shows that lower thresholds generally improve the recovery of reference peaks.

The precision graph shows the opposite trend: higher thresholds suppress additional detections and improve precision.

The false peak graph confirms that stricter thresholds reduce the number of unmatched local maxima.

The RMSE graph shows lower conditional delay errors at higher thresholds. This must be interpreted alongside the reduction in the number of detected reference components.

8.2 Delay Profile Comparison

Figure 2 presents the delay profiles at 10 dB SNR.

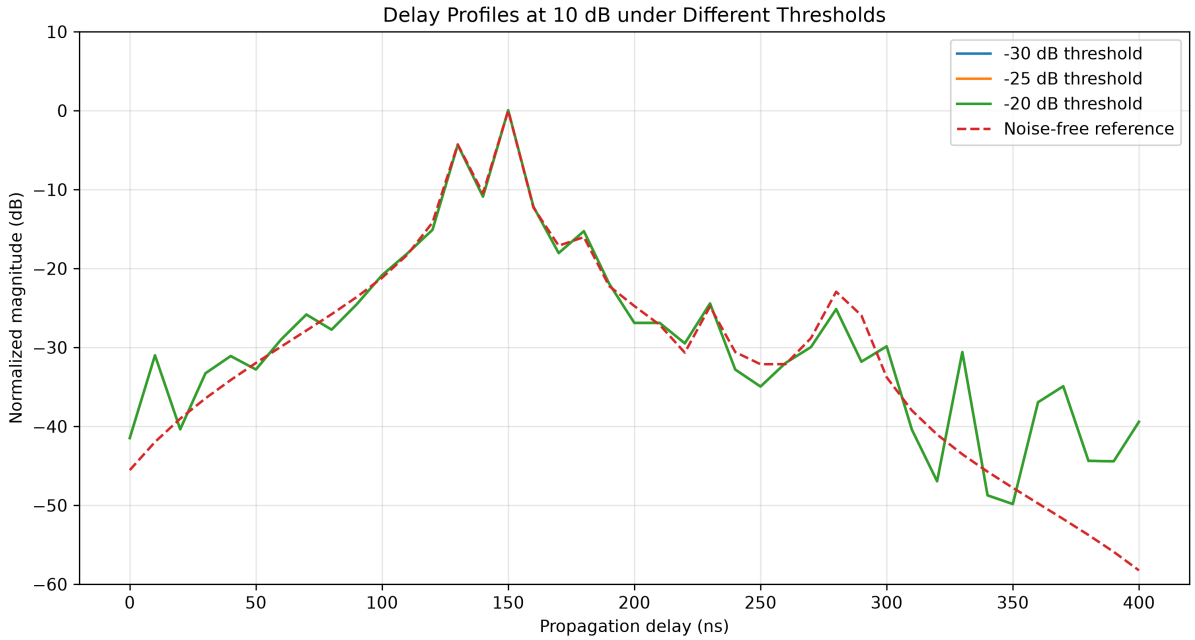


Figure 2: Delay profiles at 10 dB SNR under different threshold settings, compared with the noise-free reference profile.

The received noisy delay profile is identical for the three threshold settings within the same trial.

The threshold changes the detection decision, not the underlying received waveform.

Consequently, threshold comparison should be understood as a comparison of peak acceptance decisions rather than a change in the channel response itself.

9 Discussion and Key Findings

9.1 Detection Probability Versus Precision

The experiment demonstrates a clear trade-off between detection probability and precision.

At 20 dB, the -30 dB threshold achieves 100% detection probability, whereas the -20 dB threshold achieves only 60%.

However, the -20 dB threshold achieves 100% precision because it rejects weak components and additional unmatched detections.

Therefore, a detector must balance detection completeness against false detections.

9.2 False Detection Suppression

At -5 dB, increasing the threshold from -30 dB to -20 dB reduces false peaks per trial from 8.40 to 3.90.

The percentage reduction is:

$$\text{Reduction} = \frac{8.40 - 3.90}{8.40} \times 100 \tag{48}$$

$$= 53.57\% \tag{49}$$

This improvement comes at the expense of a reduction in detection probability from 85.8% to 74.6%.

9.3 SNR-Dependent Threshold Selection

The highest observed F1-score for each SNR condition is summarised below.

Table 6: Best observed threshold based on F1-score

SNR (dB)	Selected threshold	F1-score	Detection (%)
-5	-20 dB	0.591	74.60
0	-20 dB	0.706	61.00
5	-25 dB	0.813	84.40
10	-25 dB	0.917	91.00
15	-30 dB	0.956	99.60
20	-30 dB	0.996	100.00

The observed best-F1 threshold changes with SNR.

A candidate threshold policy suggested by these results is:

$$T(\text{SNR}) = \begin{cases} -20 \text{ dB}, & \text{SNR} \leq 0 \\ -25 \text{ dB}, & 0 < \text{SNR} \leq 10 \\ -30 \text{ dB}, & \text{SNR} > 10 \end{cases} \tag{50}$$

This is an empirical candidate policy derived from the tested threshold values. It is not yet a validated adaptive detection algorithm.

Independent validation is required before generalising this policy to other channel conditions or SNR values.

9.4 Interpretation of Delay RMSE

The lowest RMSE does not necessarily indicate the best overall detector.

At 20 dB, the -20 dB threshold achieves an RMSE of 0.1591 ns, but only 60% of reference peak opportunities are detected.

This occurs because the RMSE is calculated only over matched estimates.

A detector that rejects difficult weak components can produce a small conditional RMSE while performing poorly in terms of detection completeness.

For this reason, RMSE should always be reported alongside detection probability, precision and false peaks per trial.

10 Limitations

The following limitations apply:

1. Only three fixed detection thresholds were evaluated.
2. The reference delay positions were obtained from a noise-free simulation and retained across all threshold conditions.
3. The channel is static and does not include vehicle motion or Doppler effects.
4. Ideal synchronization is assumed.
5. The evaluation uses a single OFDM symbol per trial.
6. The delay RMSE is conditional on successful peak matching.
7. The SNR-dependent threshold policy is selected using the same simulation dataset and requires independent validation.
8. The experiment has not yet been validated using physical SDR hardware.

11 Conclusion

Experiment 16 investigated the sensitivity of OFDM-based multipath detection to the selection of detection thresholds under different SNR conditions.

Three fixed thresholds, -30 dB, -25 dB and -20 dB, were evaluated across six SNR conditions using 100 Monte Carlo trials per condition.

The results demonstrate that lower thresholds improve the detection of weak multipath components but can increase false detections.

Higher thresholds improve precision and suppress false peaks, but can reject weaker propagation components even at high SNR.

At 20 dB, the -30 dB threshold achieves 100% detection probability and an F1-score of 0.996. In comparison, the -20 dB threshold achieves 100% precision but detects only 60% of reference peak opportunities.

The experiment therefore demonstrates that a single fixed threshold is insufficient for optimising detection performance across all tested SNR conditions.

The results provide the motivation for developing an SNR-dependent adaptive threshold method in the subsequent experiment.